

HAMJETH MISREE

Machine Learning Engineer | Associate Technical Lead | Senior Software Engineer

hamjeth68@gmail.com | +94 771157106 | linkedin.com/in/hamjeth-misree | github.com/Hamjeth68 | hamjeth.vercel.app

No 40 Kawdana Road, Dehiwala – Mount Lavinia, Sri Lanka

PROFESSIONAL SUMMARY

Engineering lead and senior full-stack engineer with 6+ years of experience across web, mobile, backend, cloud-backed systems, and AI/ML engineering. Deep hands-on expertise in LLM-powered agentic architectures using LangGraph, LangChain, OpenAI GPT-4, Anthropic Claude SDK, AWS Bedrock, and Hugging Face Transformers. Proficient in RAG pipelines, Transformer architecture, cosine/sine similarity search, and HIPAA-compliant AI system design. Delivered production systems across autonomous AI health platforms, school management SaaS, transport operations, fintech, and enterprise EdTech products.

PROFESSIONAL EXPERIENCE

CDAZZDEV – Machine Learning Engineer

May 2026 – Present | Sri Lanka

Project: AI Health Platform – LLM-Powered Agentic Health Intelligence <https://healthelic.com/>

- Architecting and building an AI-powered health platform leveraging multi-agent orchestration for personalised patient guidance, symptom assessment, and intelligent health insights delivery.
- Designed a Supervisor → Specialist agent pipeline using LangGraph with stateful graph execution, human-in-the-loop interrupt nodes, and persistent cross-session memory via Mem0.
- Implemented RAG pipelines with pgvector and Pinecone using cosine and sine similarity search to ground LLM responses in verified medical knowledge bases and clinical guidelines.
- Integrated OpenAI GPT-4 and Anthropic Claude (Anthropic SDK) alongside AWS Bedrock-hosted foundation models for flexible, scalable LLM inference across health workflows.
- Leveraged Hugging Face Transformers and pre-trained biomedical models for domain-specific NLP tasks including entity extraction, classification, and semantic similarity scoring.
- Applied Transformer architecture concepts (attention mechanisms, embeddings, positional encoding) to fine-tune models on health-specific datasets, improving domain accuracy.
- Built FastAPI (Python) service layer with AWS Bedrock API integration for agent orchestration, webhook ingestion, and multi-channel routing across mobile and web surfaces.
- Designed and enforced HIPAA-compliant data architecture: per-user Redis namespacing, PostgreSQL Row-Level Security, encrypted PII at rest and in transit, and full audit logging.
- Implemented AWS Strands agent framework for structured tool-use workflows, enabling agents to safely call external health APIs and internal knowledge retrieval tools.
- Applied LLM fine-tuning strategies and prompt optimisation to reduce hallucination rates in health-critical output contexts.

Technologies: Python, FastAPI, PostgreSQL, Redis, Docker, AWS Bedrock, AWS Strands, React Native, TypeScript, AWS (EC2, S3, CloudWatch)

Jasmin Media – Associate Technical Lead

February 2025 – April 2026 | Sri Lanka

Project: Strike7 – Autonomous AI Pentesting Platform <https://strike7.ai/>

- Contributed to a platform where AI agents reason, probe, and chain exploits across domains, networks, and APIs.
- Supported end-to-end delivery for strike execution workflows, evidence capture, exploit-chain traceability, and audit-ready reporting.
- Drove maintainable engineering patterns and technical decisions across application and service layers.

Technologies: Next.js, TypeScript, AI Agent Workflows, REST APIs, Cloud-Native Architecture, AWS

Project: TMS – Transport Management System <https://service.felcon7.com/login>

- Contributed to a transport management platform spanning dispatcher dashboards, reservation execution workflows, and driver-facing application flows.
- Worked across reservation lifecycle behaviour, dispatch queue support, real-time updates, trip-state progression, and notification-driven operational visibility.
- Supported end-to-end implementation across frontend, backend, state, and API contract layers while preserving production-safe workflow integrity.

Technologies: React.js, React Native, TypeScript, NestJS, PostgreSQL, WebSocket, Firebase Cloud Messaging (FCM), REST APIs, Docker

PelicanCube PVT Limited – Senior Software Engineer – Full Stack

January 2025 – August 2025 | Sri Lanka

Project: Pravesha – Sri Lanka Railways E-Ticketing Platform <https://pravesha.lk/sn/eticket>

- Led development of a high-volume public digital ticketing platform.
- Architected scalable mobile and web solutions with multilingual support and real-time booking workflows.
- Implemented secure QR-based e-ticketing and payment flows while improving delivery quality across the team.

Technologies: React.js, React Native, TypeScript, Tailwind CSS, MMKV, React Query, Redux Persist, NestJS, Redis, PostgreSQL, Docker, NGINX, AWS (EC2, S3, CloudWatch)

Sanlabz PVT Limited – Senior Software Engineer

January 2024 – January 2025 | Sri Lanka

Project: ClassVeew – SaaS Platform for Preschools & Kindergartens (Qatar)

- Led full-stack delivery of a school management platform across mobile and web.
- Implemented offline-first behaviour, real-time updates, payments, attendance, and parent communication workflows.

Technologies: React.js, React Native, TypeScript, Tailwind CSS, MMKV, React Query, Redux Persist, NestJS, Redis, PostgreSQL, Docker, NGINX, AWS, GCP, GitHub Actions, Fastlane, Firebase, OpenAI, RabbitMQ, Prisma, TypeORM

Codelantic (Pvt) Limited – Software Engineer

September 2021 – December 2023 | Sri Lanka

Project: SpotonMoney – Cross-Platform Remittance Product <https://classic.spoton.money/>

- Built cross-platform remittance product with onboarding, recipient management, and transfer tracking.

Technologies: React Native, TypeScript, Redux, REST APIs, Secure Document Upload, Fintech Integrations

Project: Bankyo Compliance – Modular Banking Ecosystem

- Developed modular banking ecosystem with secure authentication and API-driven integrations.

Technologies: React.js, TypeScript, Node.js, Microservices, Docker, Kubernetes, REST APIs

Project: Restaurant & Hospitality Platforms

- Delivered multiple hospitality platforms including Namasthe India, RestaurantX, and Trinco Foodie Park.

Technologies: React.js, React Native, TypeScript, Redux, Spring Boot, NestJS, PostgreSQL, Docker, Payment Gateway Integrations

Koombiyo IT – Software Engineer (Intern)

January 2019 – September 2021 | Sri Lanka

Project: KoombiyoDelivery.lk – Logistics & Delivery Platform

- Worked across tracking, notifications, and logistics workflows for a delivery platform.

Technologies: React.js, Node.js, Redux, Firebase, MongoDB, REST APIs, WebSocket

Project: Ayp.lk – Buy & Sell Marketplace

- Built responsive React-based marketplace experiences for buy-and-sell flows.

Technologies: React.js, Node.js, Redux, Firebase, MongoDB, Azure DevOps, Express.js, Twilio

TECHNICAL SKILLS

AI / ML Engineering: LangGraph, LangChain, AWS Strands, OpenAI GPT-4, Anthropic Claude SDK, AWS Bedrock, Hugging Face Transformers, Transformer Architecture, RAG (Retrieval-Augmented Generation), Vector Databases (pgvector, Pinecone), Cosine Similarity, Sine Similarity, LLM Fine-Tuning, Prompt Engineering, Multi-Agent Orchestration, HITL (Human-in-the-Loop), Semantic Embeddings, Mem0

Frontend: React.js, Next.js, Angular, TypeScript, JavaScript (ES2022+), Redux, Redux Persist, React Query, HTML5, CSS3, Tailwind CSS, Shadcn UI

Mobile: React Native, Flutter, Expo, Android (Java / Kotlin), iOS (Swift / Objective-C), MMKV

Backend: Node.js, NestJS, FastAPI (Python), Spring Boot, ASP.NET, GraphQL, REST APIs, WebSocket, Microservices, RabbitMQ, Celery, Prisma, TypeORM

Databases: PostgreSQL (+ pgvector), MySQL, MongoDB, Firebase Firestore, Redis, Pinecone

Cloud & DevOps: AWS (EC2, S3, RDS, Bedrock, CloudWatch), GCP, Azure, Docker, Kubernetes, NGINX, CI/CD, GitHub Actions, Fastlane

Security & Compliance: HIPAA Compliance, Data Encryption (at rest & in transit), Audit Logging, PostgreSQL Row-Level Security, OAuth2, JWT, Multi-Tenant Isolation

Testing & Methodology: TDD, Agile / Scrum, E2E Testing, Consumer-Driven Contract Testing, UML

Integrations: Stripe, Twilio, SendGrid, Firebase Cloud Messaging (FCM), Sentry, Firebase Analytics

EDUCATION

BSc (Hons) Computer Science – Specialisation in Software Engineering

Kingston University London · 2021

Higher National Diploma in Software Engineering

Pearson Assured, ESoft Metro Campus, Colombo 04 · 2019